

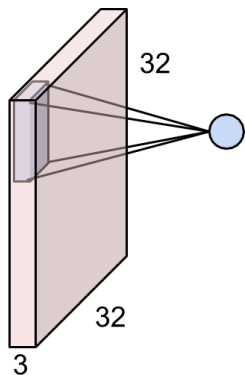
AIAA 2205: Introduction to AI

Enyan Dai

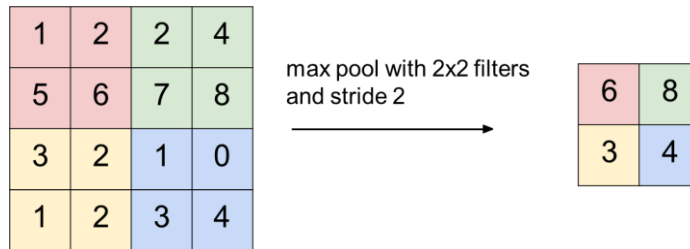
The Hong Kong University of Science and Technology (Guangzhou)

Recap: More Components of CNNs

CNN layer



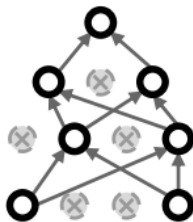
Pooling layer



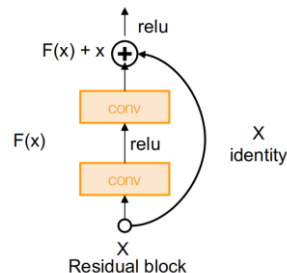
Normalization Layers

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \varepsilon}}$$

Dropout

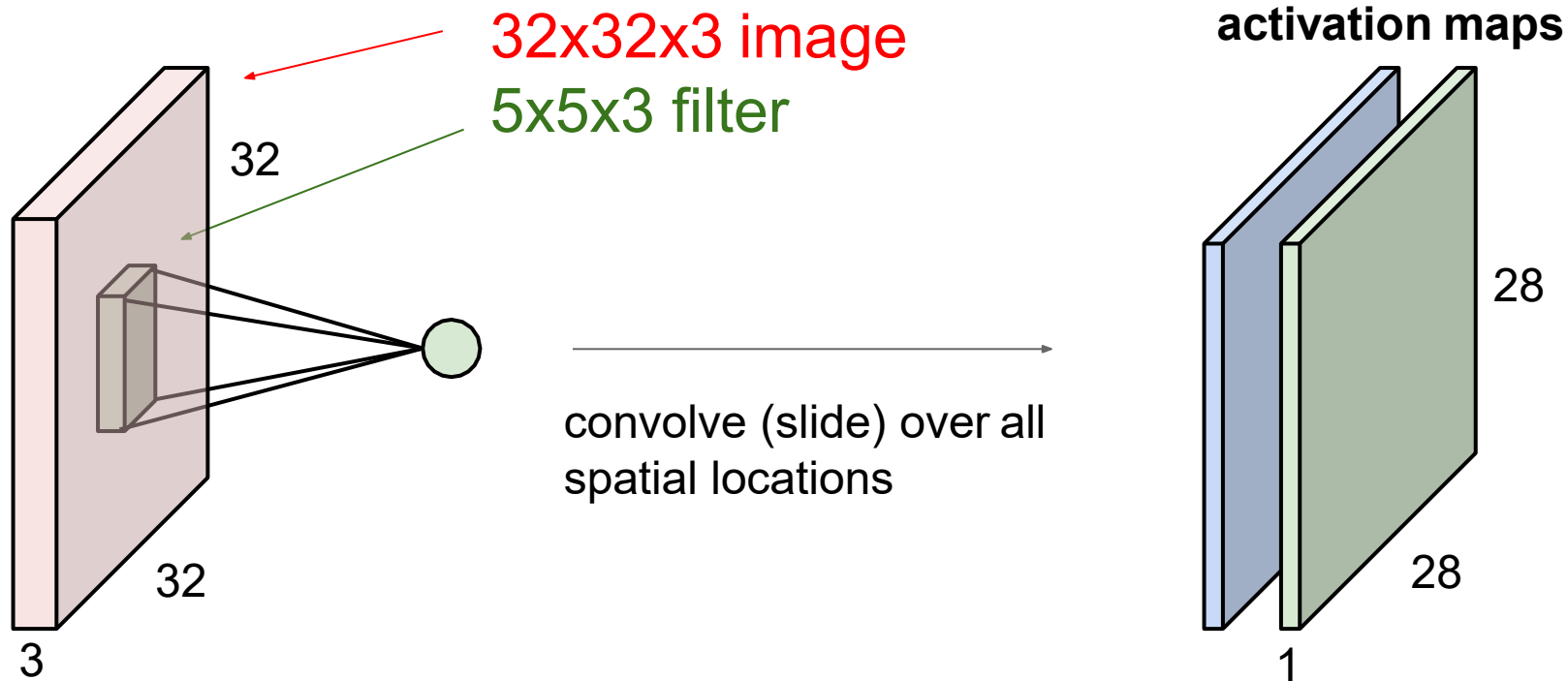


Residual Connections



Convolution Layer

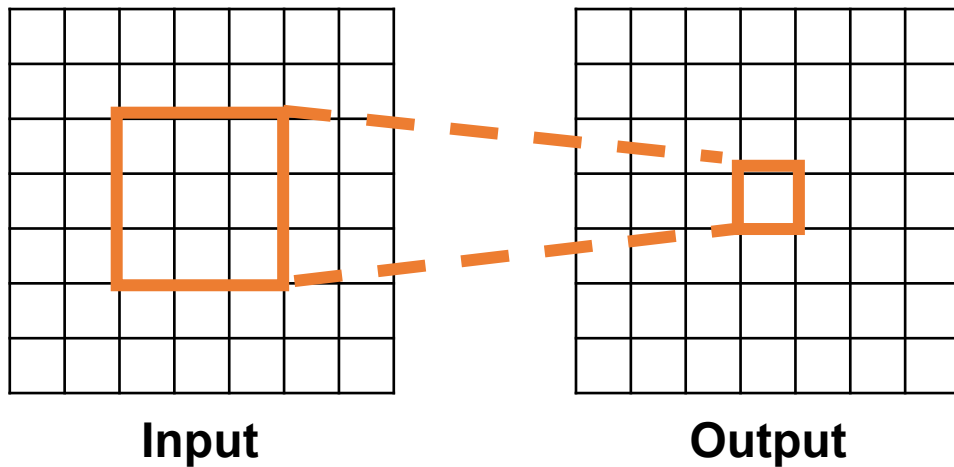
consider a second, **green** filter



Receptive field (感受野)

For convolution with **kernel size K**, each element in the output depends on a $K \times K$ receptive field in the input

kernel size 3



Max pooling

1	2	2	4
5	6	7	8
3	2	1	0
1	2	3	4

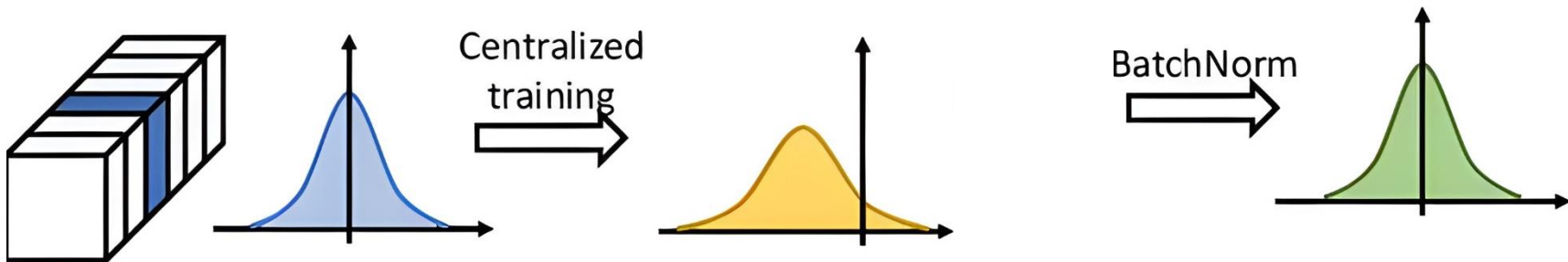
max pool with 2x2 filters
and stride 2



6	8
3	4

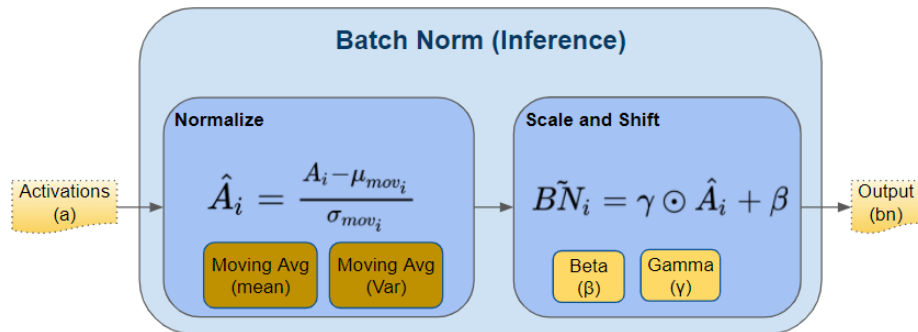
Normalization Layer

Internal covariate shift



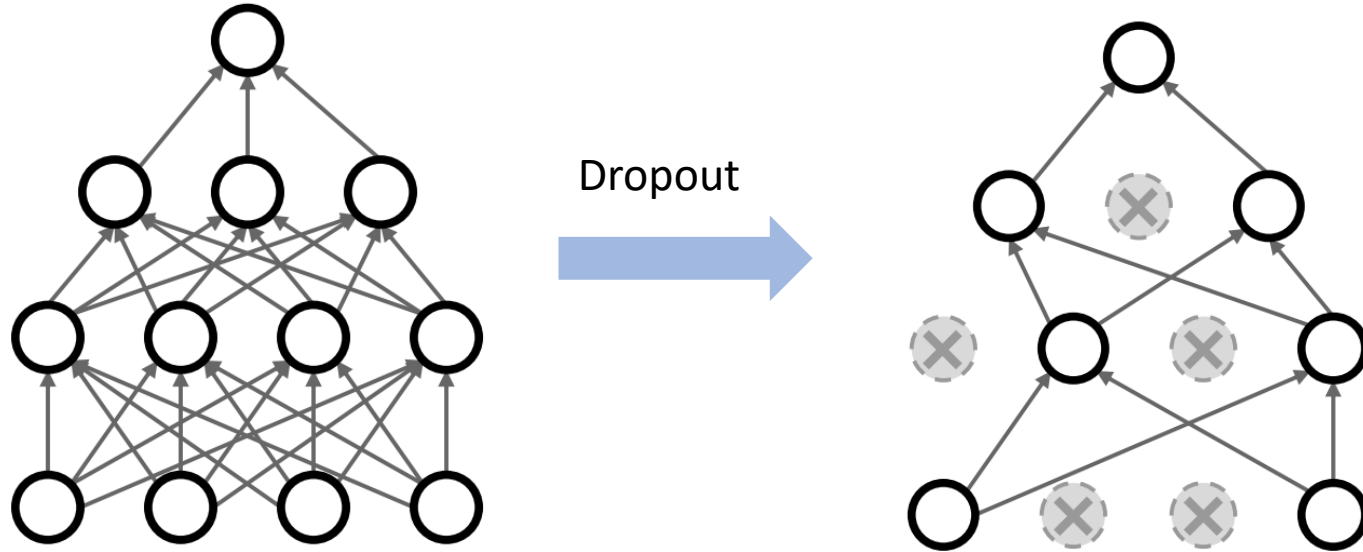
Solution: Learn parameters that let us scale / shift the input data

1. Normalize input data
2. Scale / shift using learned parameters



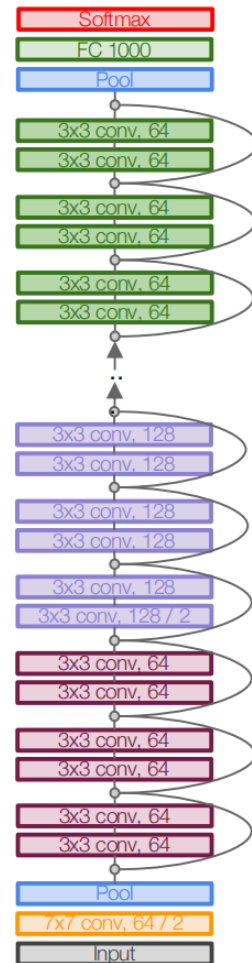
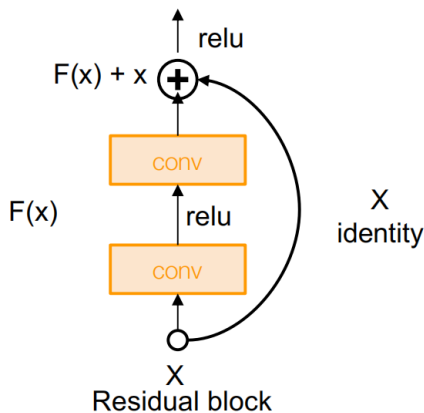
Regularization: Dropout

In each forward pass, randomly set some neurons to zero
Probability of dropping is a hyperparameter



ResNet

- **152-layer model** for ImageNet
- **ILSVRC'15 classification winner (3.57% top 5 error)**
- **Win all classification and detection competitions** in ILSVRC'15 and COCO'15!



Computer Vision Tasks

Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

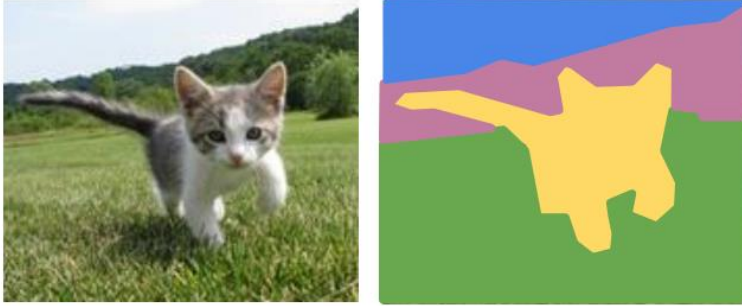
Instance Segmentation



DOG, DOG, CAT

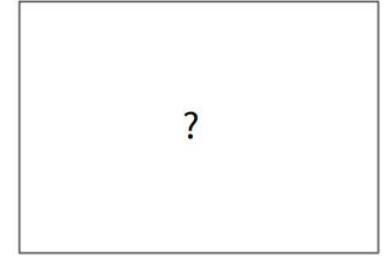
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Semantic Segmentation



GRASS, CAT, TREE,
SKY, ...

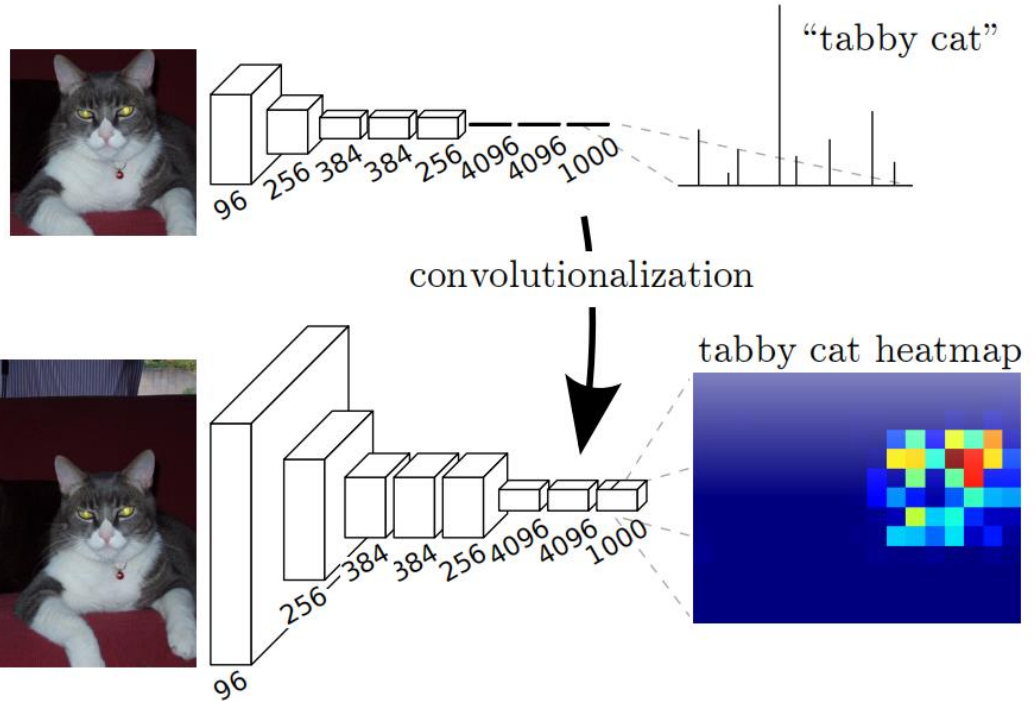
Paired training data: for each training image,
each pixel is labeled with a semantic category.



At test time, classify each pixel of a new image.

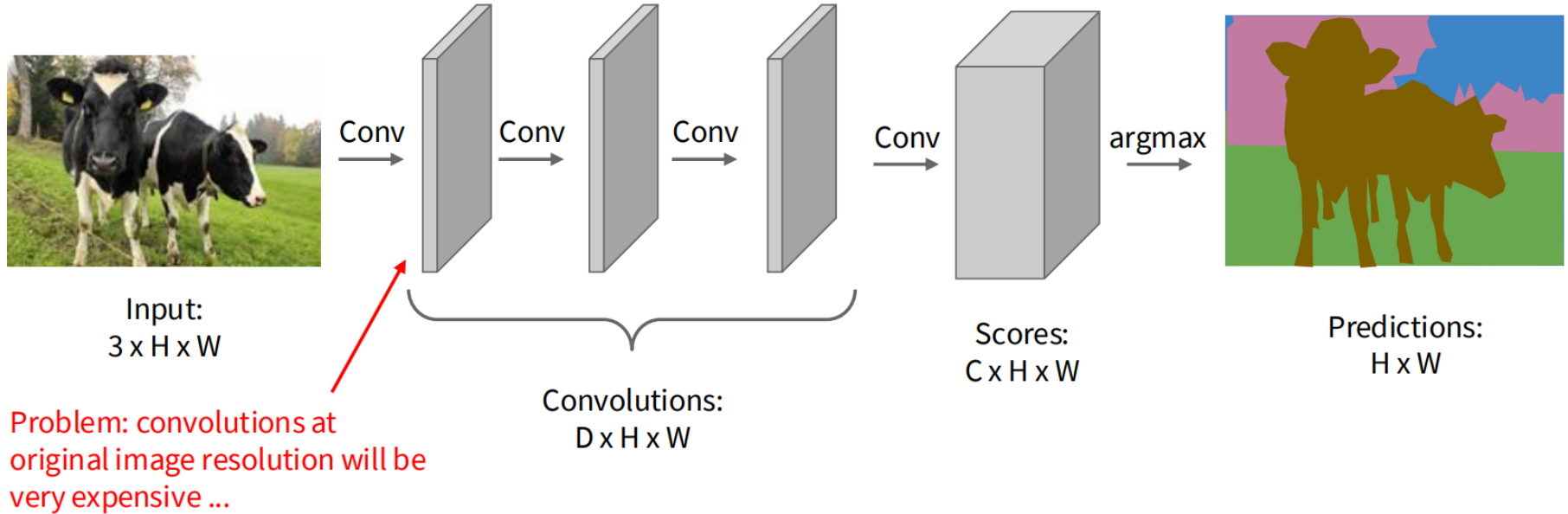
High-level Idea

We can slide the window



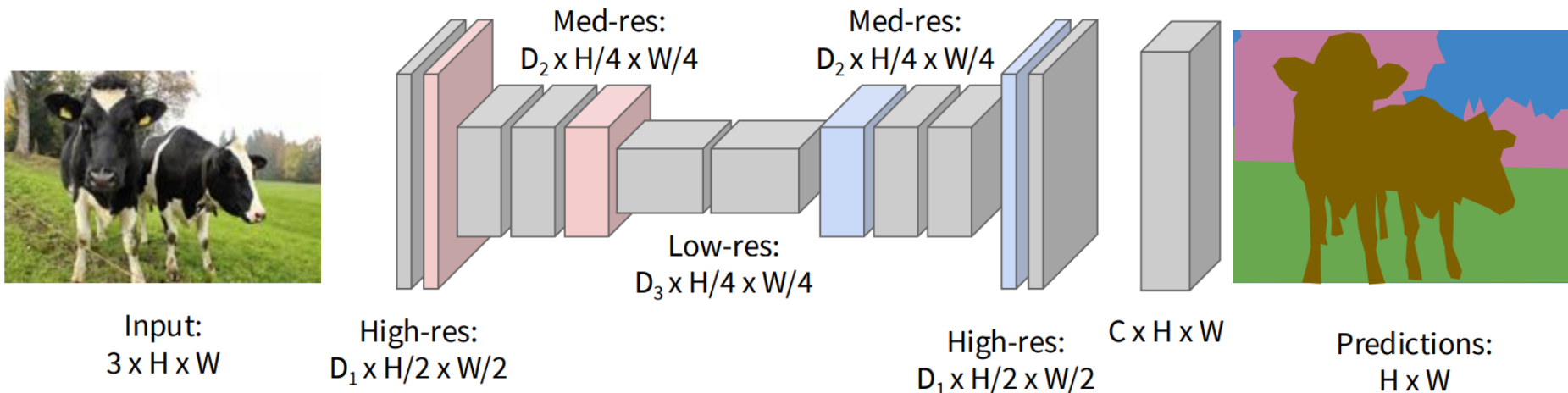
High-level Idea

Only use convolutional layers without downsampling layers



Downsampling and Upsampling

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Upsampling

Nearest Neighbor

1	2
3	4



1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Input: 2 x 2

Output: 4 x 4

"Bed of Nails"

1	2
3	4

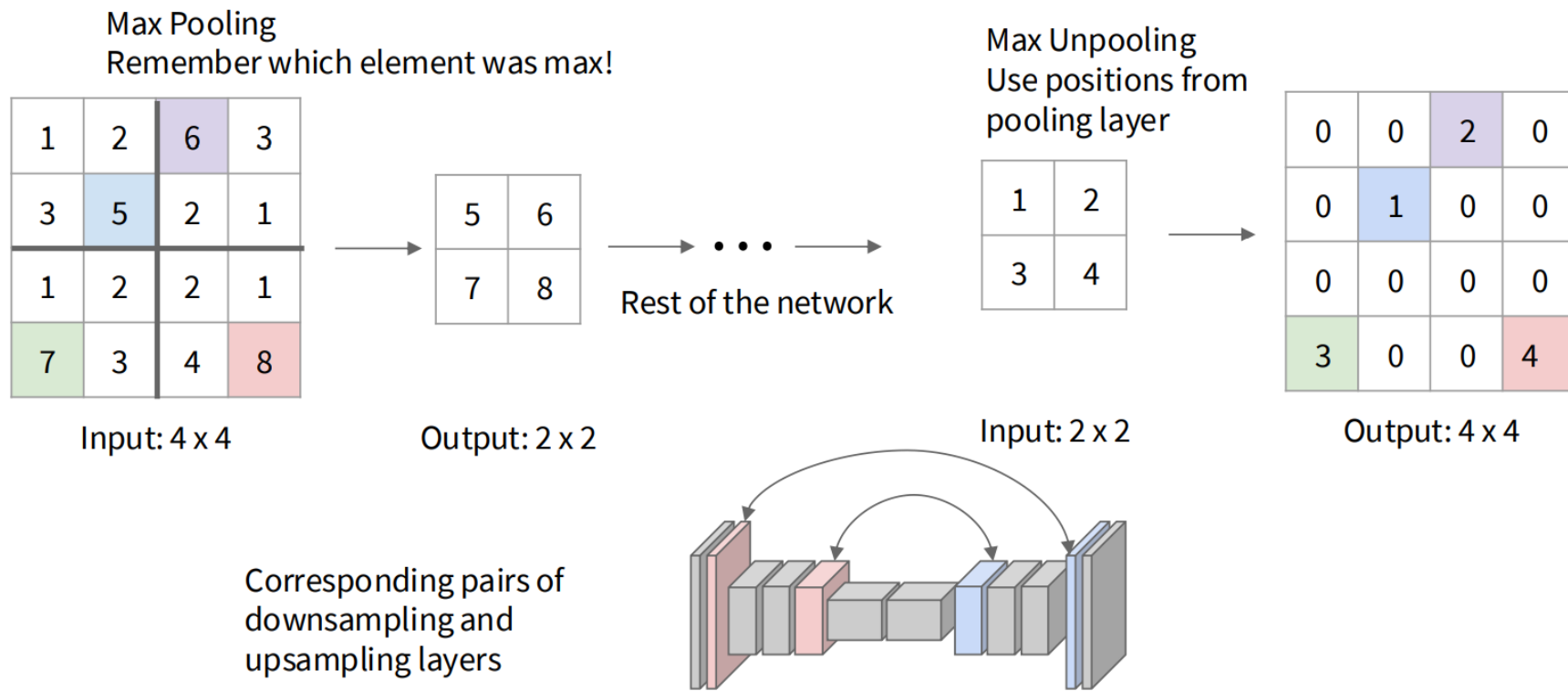


1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

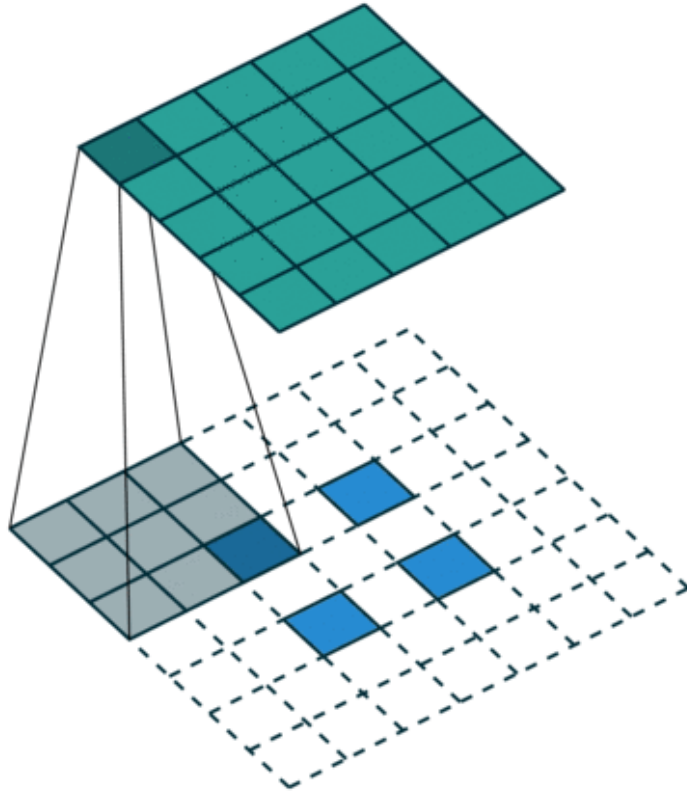
Input: 2 x 2

Output: 4 x 4

Upsampling

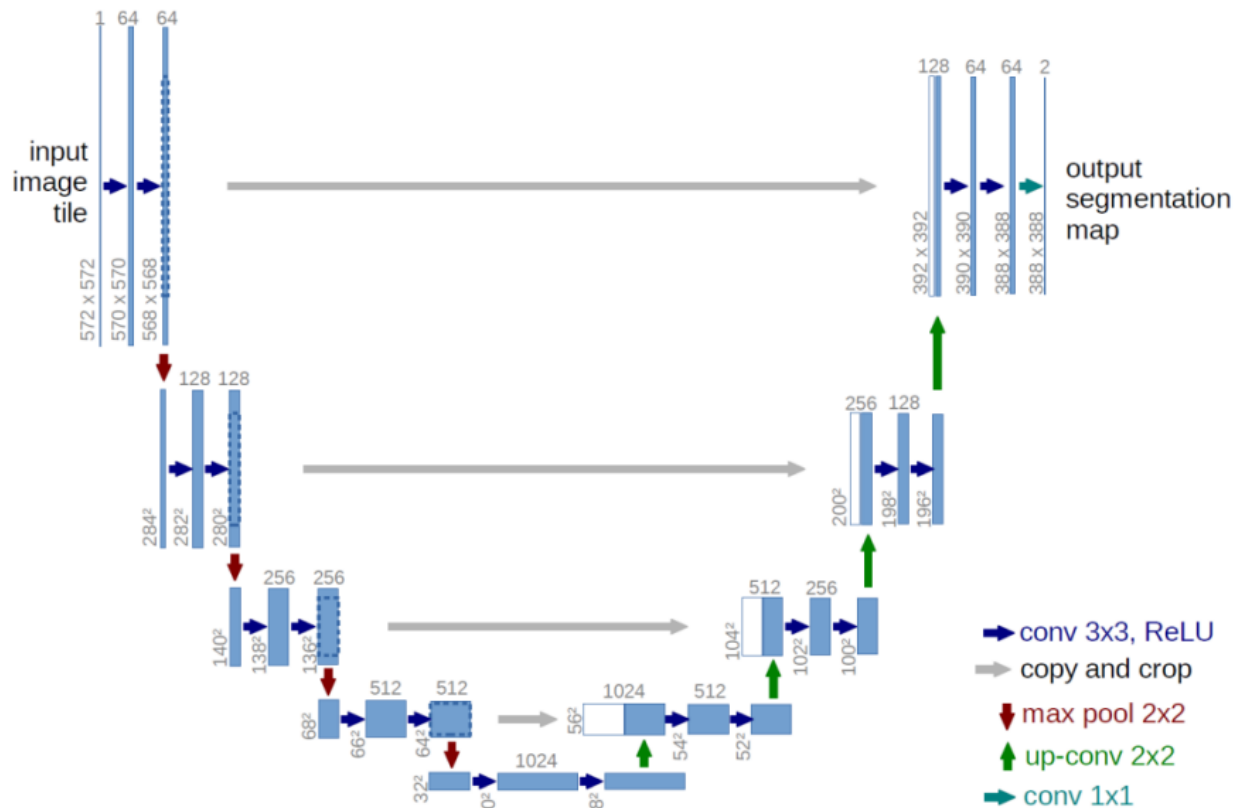


Upsampling

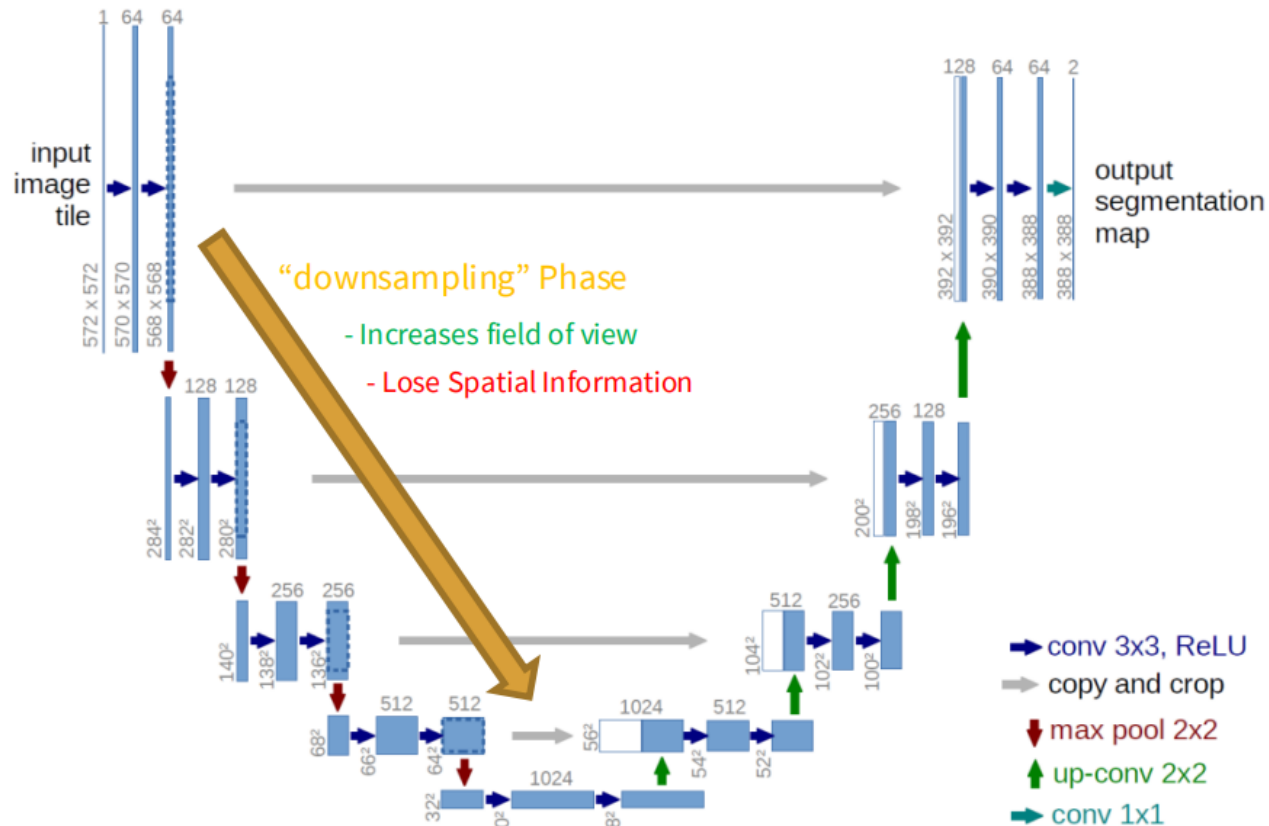


Transpose Convolution

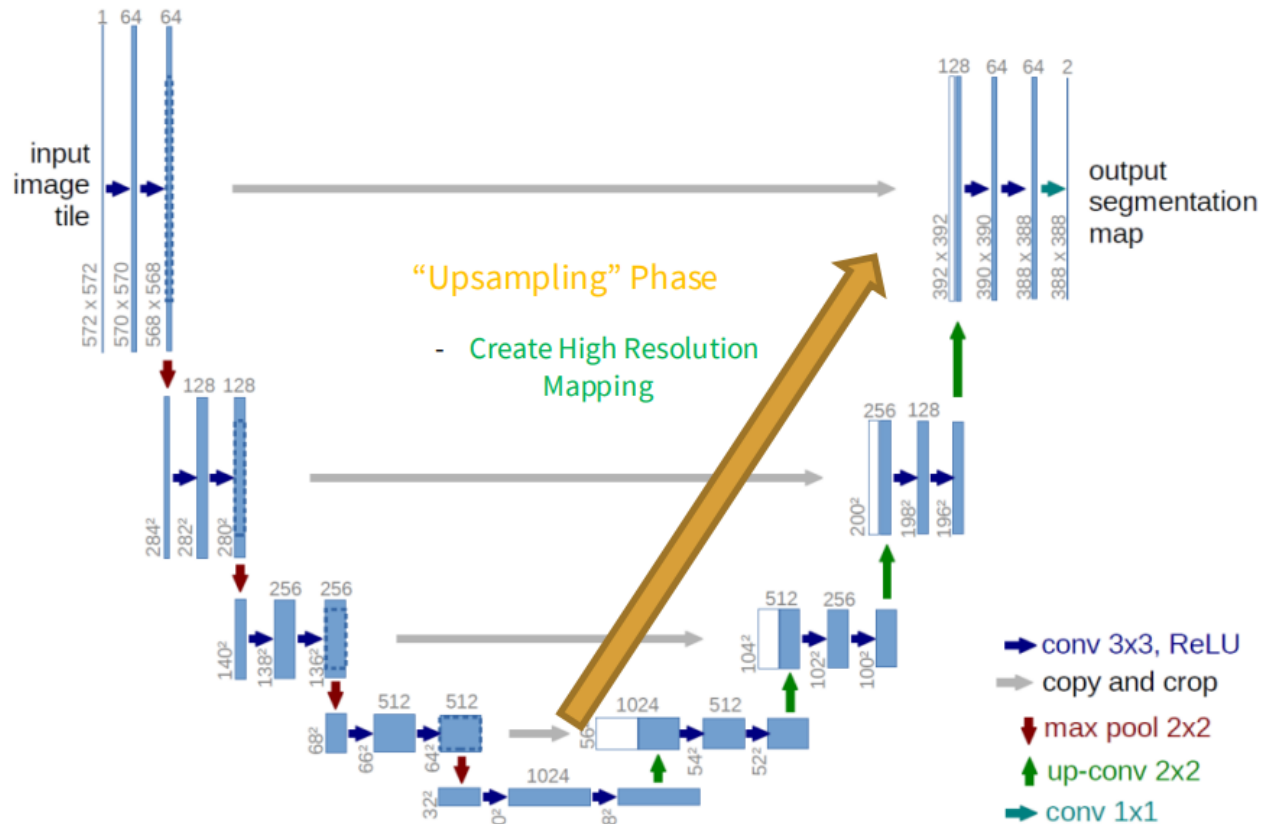
U-Net



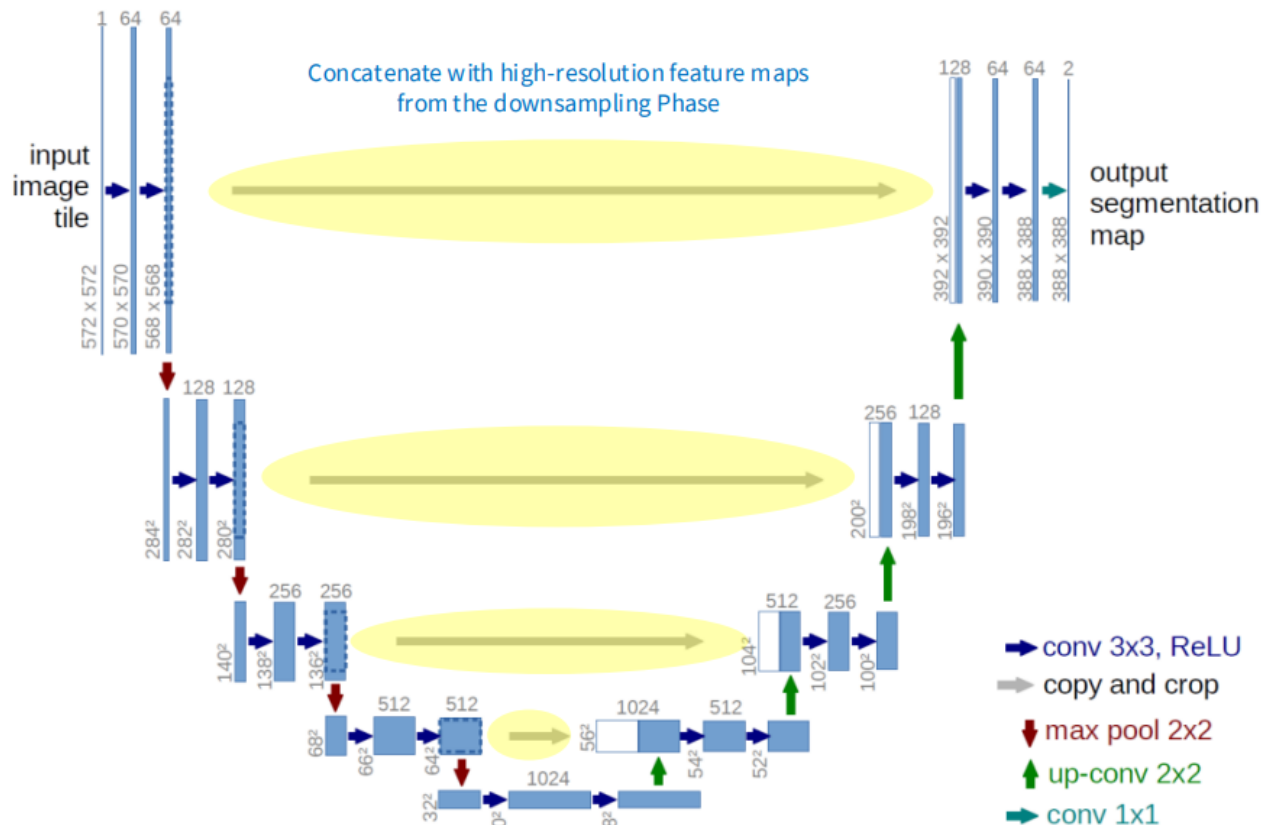
U-Net



U-Net



U-Net





近年来

引入

人脸识别是什么

人脸识别系统的组成及原理

结尾

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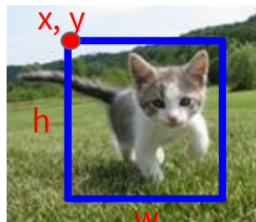


DOG, DOG, CAT

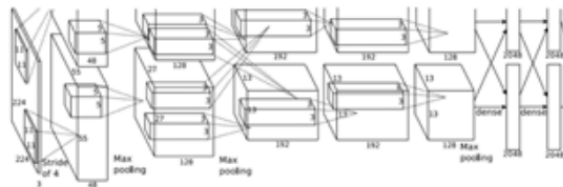
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Object Detection

Object Detection: Single Object (Classification + Localization)



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Fully
Connected:
4096 to 1000

Class Scores

Cat: 0.9
Dog: 0.05
Car: 0.01
...

Correct label:
Cat

Softmax
Loss

Multitask Loss

Vector:
4096

Fully
Connected:
4096 to 4

Box
Coordinates
(x, y, w, h)

+

Loss

L2 Loss

Correct box:
(x', y', w', h')

Treat localization as a
regression problem!

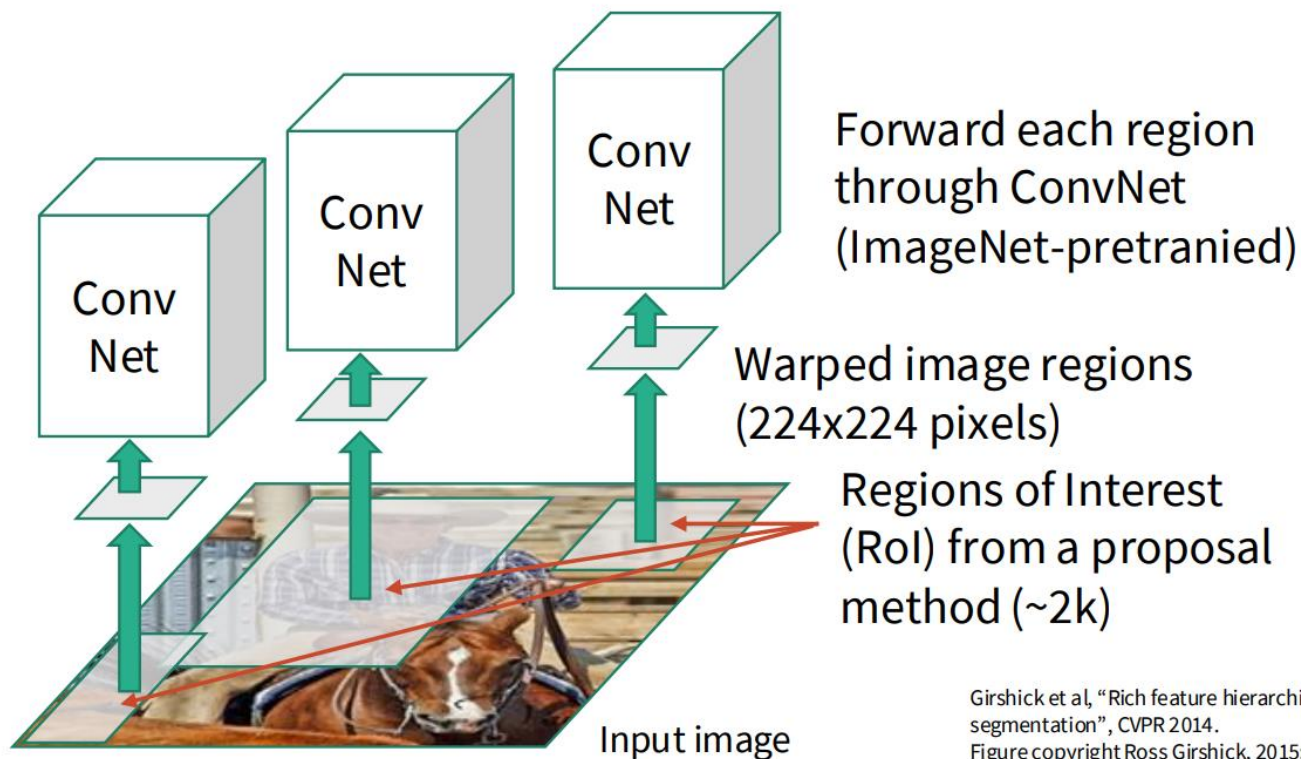
R-CNN



Regions of Interest
(RoI) from a proposal
method (~2k)

Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.
Figure copyright Ross Girshick, 2015; [source](#). Reproduced with permission.

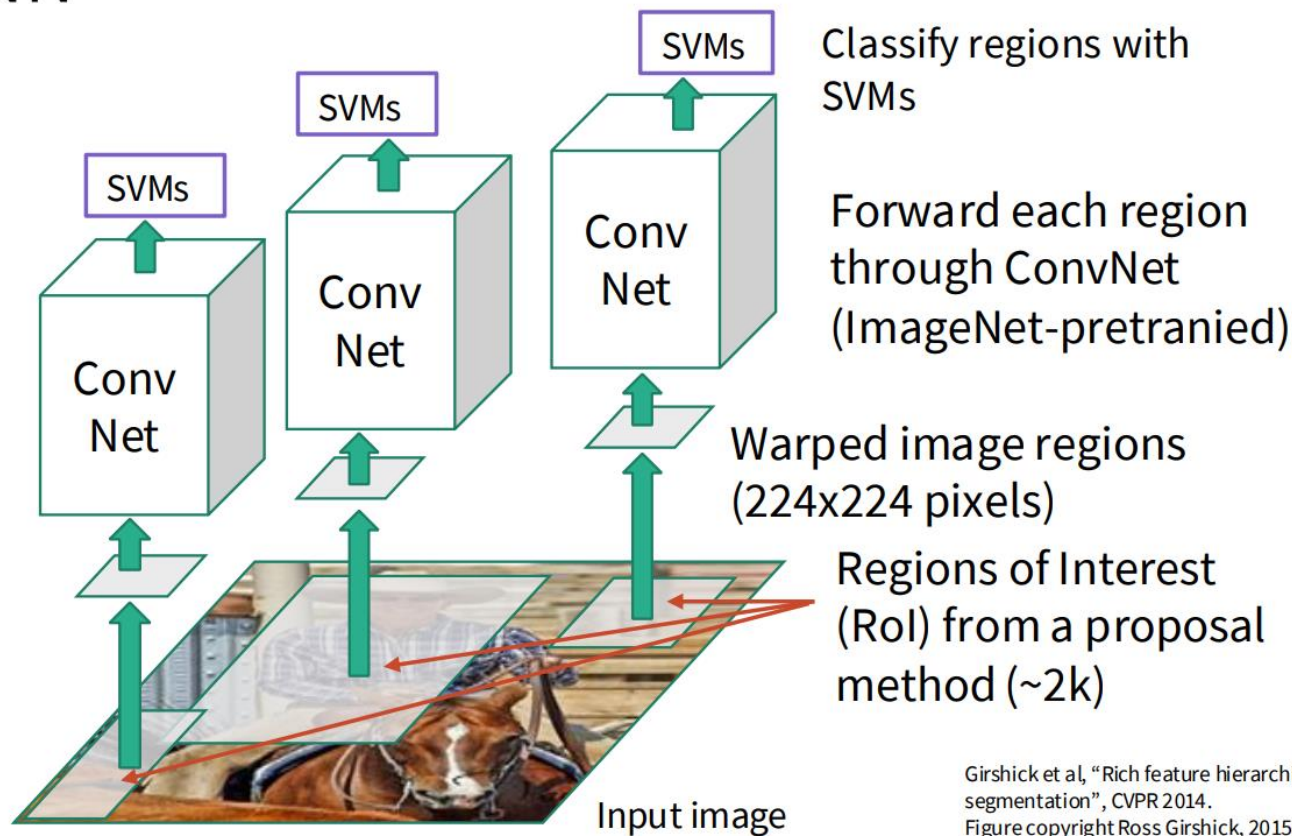
R-CNN



Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.

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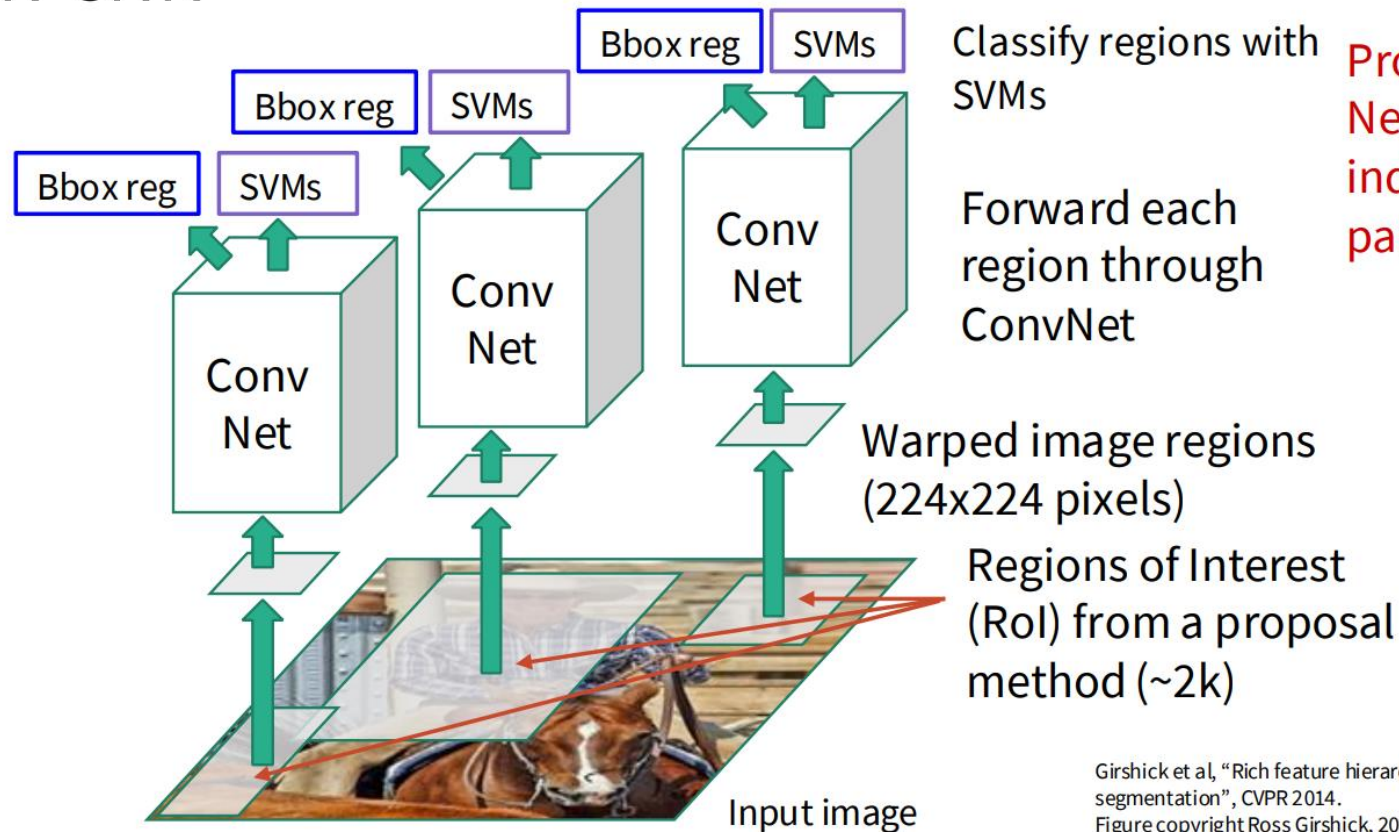
R-CNN



Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.

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R-CNN



Predict “corrections” to the RoI: 4 numbers: (dx, dy, dw, dh)

Classify regions with SVMs

Forward each region through ConvNet

Problem: Very slow!
Need to do ~2k independent forward passes for each image!

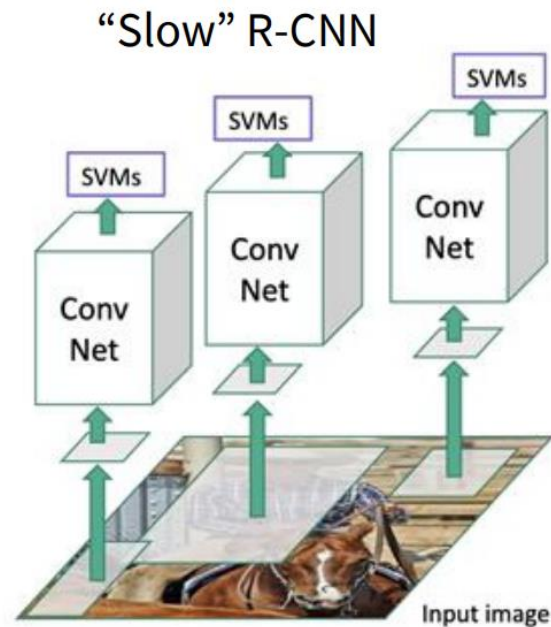
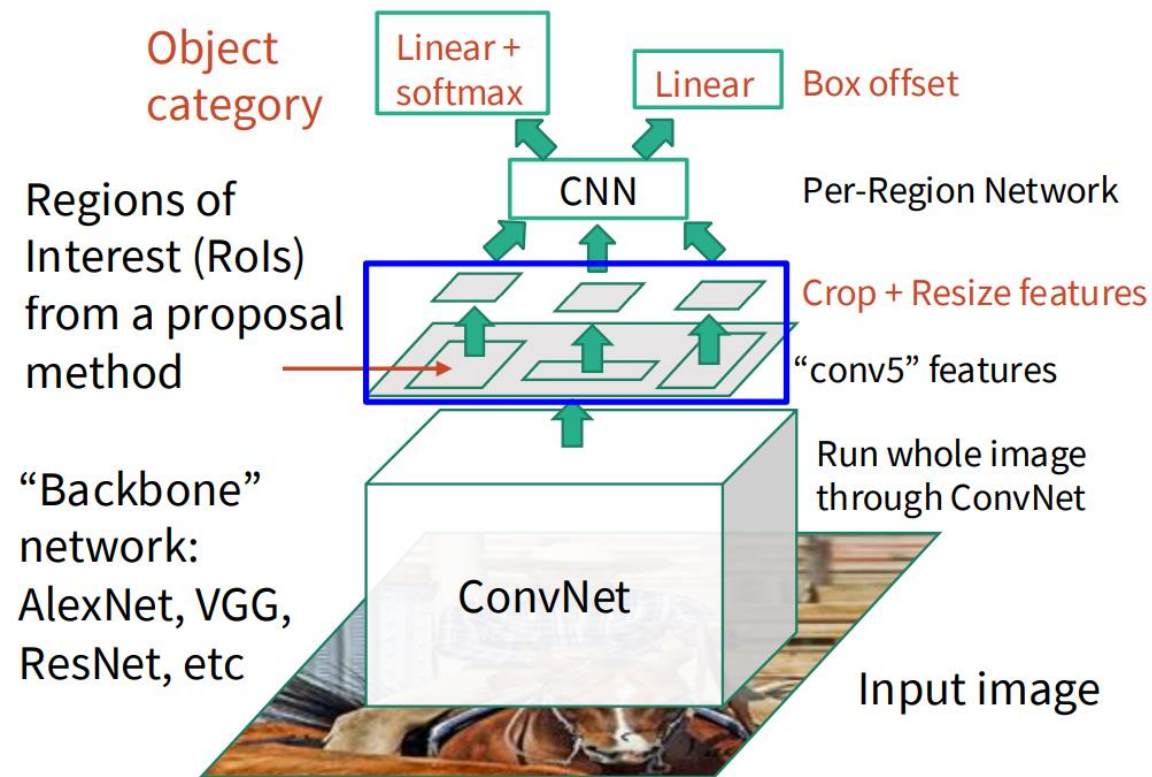
Warped image regions (224x224 pixels)

Regions of Interest (RoI) from a proposal method (~2k)

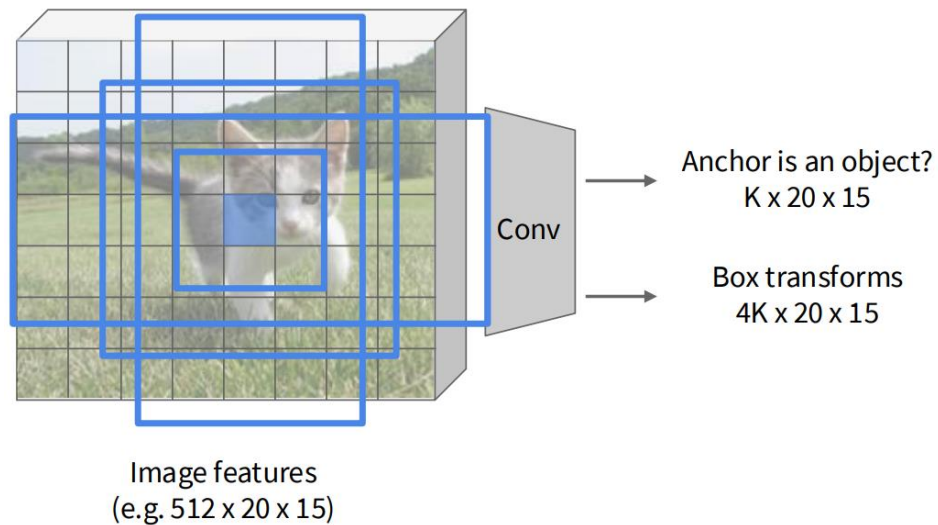
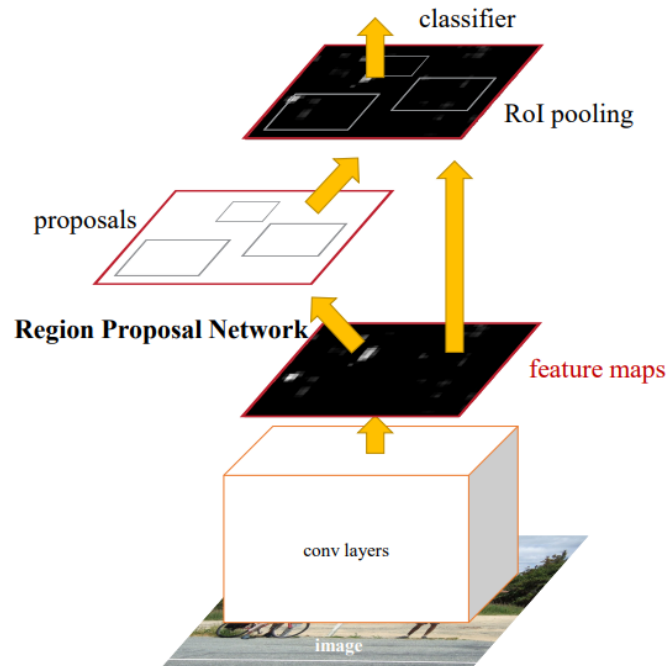
Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014.

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Fast R-CNN



Faster RCNN deploys a region proposal network



Applications
